

PHYS 211 – Computational Homework #2

Hydrogen Atom

Objective:

In this assignment, you will numerically investigate and compare the quantum behavior of particles confined in infinite and finite square wells. You will explore quantized energies, wave functions, probability densities, tunneling, and the role of boundary conditions in quantum mechanics. 20 points for each problem.

You may use MATLAB, Mathematica, or Python. You can use ChatGPT if you need help with the coding, but please disclose it in your submission.

Problem 1 – Calculate H energy levels for $n=1, 2, 3, 4, 5, 6$ in eV.

Problem 2 – Plot Coulomb Potential ($U(r)$) from $r=0.01$ nm to $r=2$ nm. Convert energy axis to eV.

Problem 3 – Calculate Bohr radius in Angstroms.

Problem 4 – Plot H atom radial probability distributions for 1s (ground state), 2s, and 3s states.

Problem 5 – Use spherical harmonics to visualize s, p, and d orbitals. You can use polar, density, or 3D surface plots.

Submission Requirements

- Your code
- All plots
- Numerical calculations
- Written explanations

Constants

- $h = 6.626 \times 10^{-34}$ J·s
- $\hbar = 1.055 \times 10^{-34}$ J·s
- $m_e = 9.11 \times 10^{-31}$ kg
- $1 \text{ eV} = 1.602 \times 10^{-19}$ J
- $c = 3.00 \times 10^8$ m/s

Notes

Clearly label all graphs and include axes with units. Include short interpretations and comments in your code explaining major steps.